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REVIEW

Deep Learning Applications in Textile Industry: A Systematic Review of Current Status and Delineating Future Research Agenda

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Abstract
Deep learning (DL) methods have opened unprecedented opportunities in the textile industry by providing critical insights into fibre-yarn-fabric defect detection, quality control, quality management, process automation and optimisation. The present work aims to summarise the literature related to DL applications in the textile industry and highlight future research agenda. This review combines bibliometric and network analyses of the relevant research papers. Content analysis was also done to elucidate various DL algorithms used at different stages of textile manufacturing and quality management. Convolutional neural network (CNN), deep neural network (DNN) and You Only Look Once (YOLO) are the most widely used DL techniques for fibre classification, fabric defect detection, garment classification, body shape reconstruction, etc. Some practical difficulties related to the applications of DL include nature of the data, size and scope of the data, selection of appropriate models, process uncertainty, etc. This research makes threefold contributions. First, mapping of different DL algorithms used in the textile industry; second, classifying the applications of DL based on the stages of textile manufacturing and quality management; and finally, identifying the challenges and themes for future research directions. This review provides a comprehensive reference for industries and academia on DL applications in the textile industry.

Keywords Convolutional neural network · Deep learning · Defect detection, quality management · Textile industry

1 Introduction

There are many exciting applications and ongoing research in the field of artificial intelligence (AI) [3]. AI-supported software and tools can recognise speech or images, make medical diagnoses, automate everyday tasks, and support scientific studies. An AI system that can extract patterns, without human intervention, from a large amount of unstructured data is referred to as machine learning or ML [24]. The introduction of various effective learning algorithms and network architectures made artificial neural networks (ANNs) a prominent tool of machine learning in the late 1980s [91]. Back-propagation learning algorithms, self-organising maps, and radial basis function networks (RBFNs) are some of the unique methods used to train multilayer perceptron (MLP) networks [26, 27]. Even after being fairly successful in several applications, interest in ANNs waned a bit in recent years. More studies are now being done on Kernel approaches and Bayesian graphical models, which were not focussed previously [45]. In 2006, a seminal work was published outlining the principles of deep learning [28] which led to the revival of the ANN research in the field of machine learning. ANN with more than three layers forms the basis for deep learning (DL). In recent years, DL networks have obtained unmatched performances in several classifications as well as regression problems when trained appropriately [45]. DL is increasingly becoming a key tool for automation in industrial applications. For this reason, several attempts have been made to review the extent of the work done with DL in various fields such as in pharmacy [50, 104], food processing [69, 102], chemical processing [103, 114], etc., including the textile industry.

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There are many approaches to literature review, namely narrative review, descriptive review, scoping review, systematic review, bibliometric analysis, and Meta-analysis [19]. Narrative review merely states the findings of research articles whereas descriptive review goes one step further by identifying the common patterns or trends in literature. A scoping review encompasses the length and breadth of the topic from all possible aspects and bibliometric analysis focuses on the data and statistics related to the published literature on a topic. The systematic review is a well-organised and methodical approach that requires a significant amount of data but is highly beneficial for assessing the gathered information and comprehending research patterns in the field of interest [36]. It answers some specific research questions pertinent to the topic of interest.

Textile and clothing are one of the most important industries in developing worlds including India [70]. The value chain of this industry is quite long encompassing fibre manufacturing, yarn production, fabric making, wet processing, and apparel manufacturing [7]. As textile and clothing manufacturing is highly labour intensive, the processes are susceptible to generating defects causing internal and external failure costs [74]. With the wealth of data generated at various stages of textile operations, it is impossible to manually identify patterns and trends in the data [7]. To cope with this problem, AI, ML, and DL offer an immense scope of applications in textile processes, quality management, and decision-making [84]. DL is fruitful in many areas such as fibre identification [125], fibre bundle image enhancement [94], yarn quality detection [110, 113], fabric defect detection [38, 42, 56, 79, 134], garment recognition and classification [1, 15, 18], garment shape, design, and fitting [10, 109, 111], traceability by generating coding tag in the textile supply chain [53, 109], etc. These applications are a clear indication of the growing interest in DL applications in the textile sector. A comprehensive summary of image vision applications for fabric defect identification was presented by Rasheed et al. [85]. However, a systematic review of DL applications covering the entire value chain of the textile industry is conspicuously missing. To fill this lacuna, the present work explores DL applications in textile operations and quality management. This will allow us to assess the current state of the field and create a path for future research. Accordingly, the study attempts to address the following research questions (*RQ*s):

RQ1. What are the different types of DL algorithms used in textile operations and quality management?

RQ2. What are the stages of textile operations and quality management where the DL algorithm has been used?

RQ3. What is the future direction of DL research in textile operations and quality management?

The rest of the manuscript is organised as follows: Sect. 2 presents the methodology for carrying out the literature review. Section 3 provides an overview of DL and the content analysis is provided in Sect. 4. Section 5 summarises the future research direction in the reviewed field and the concluding remarks are made in Sect. 6.

2 Methodology

This study used a three-pronged mixed approach, combining bibliometric, network, and content analyses to perform a comprehensive literature review of DL applications in textile operations and production (Fig. 1). The five-step methodology developed by Zupic and Čater [135] for conducting systematic reviews served as inspiration for this process. The purpose of the bibliometric application was to decipher the statistical trends of publications in this research area. The network analysis was employed in this study to analyse the recurring themes of research and spot promising new directions. Finally, the content analysis led to a deeper understanding of the widespread applications of various DL algorithms in various phases of operations of the textile industry.

2.1 Literature Collection

Firstly, it was required to decide on a set of keywords that would accurately describe the study's topic. According to the research of Barsky and Bar-Ilan [8], searching requires more time and energy if keywords are inappropriate. Therefore, many keywords were assessed to ensure that all relevant publications were included within the scope of the study. The search string used for the shortlisting of articles was ("deep learning" AND (textile OR fibre OR cloth* OR garment OR fabric OR yarn)). The string was searched in the title, keywords, and abstract platform.

A vast array of online databases, including Scopus, Web of Science, EBSCO Host, IEEE Xplore, Google Scholar, Science Direct, Emerald, ProQuest, and many more, are available which provide the body of knowledge on any given research topic. Scopus and Web of Science databases were used in this research for searching the articles containing the above-mentioned search string. Scopus and Web of Science offer very wide coverage among the scientific databases available [96]. As can be seen in Fig. 1, a filter was used to restrict the database to only Englishlanguage scholarly journals. To narrow the research focus, only research articles were considered and other sources such as book chapters,

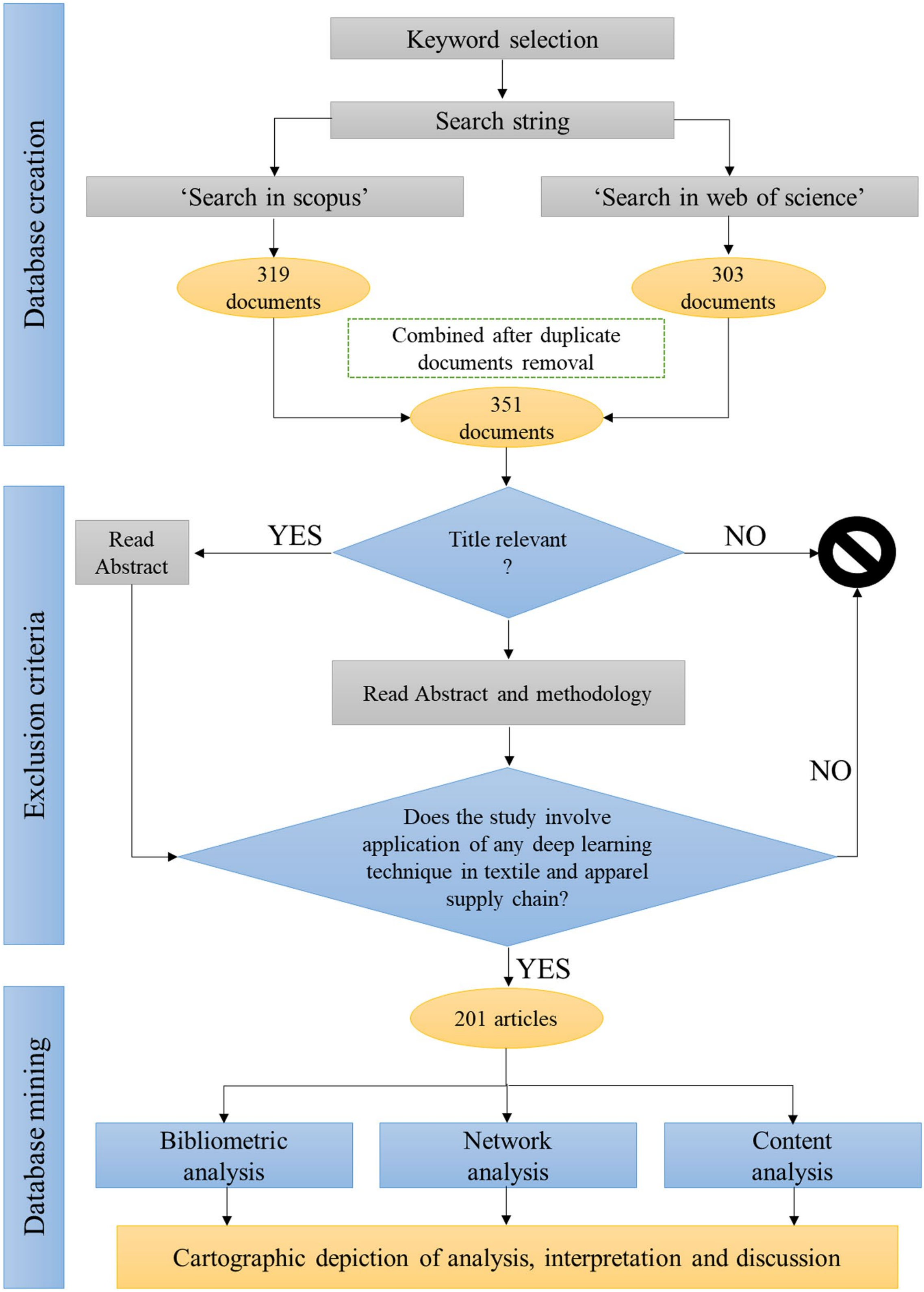

Fig. 1 Flowchart of the literature review process

textbooks, and conference proceedings were not included in this study [16, 75]. Initially, 319 and 303 articles were found in Scopus and Web of Science databases, respectively. After manual sorting and removal of duplicate articles common

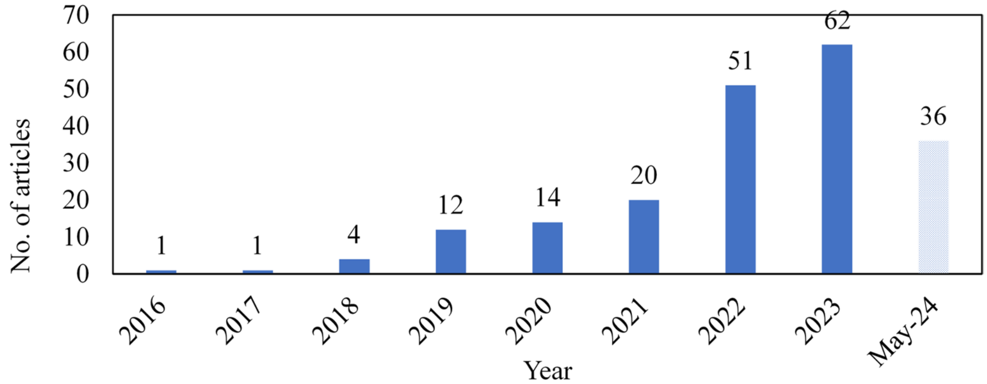

Fig. 2 Publication trend over the years

Table 1 Different bibliographic metrics of the reviewed articles

Publication Information	Results
Total number of publications	201
Total number of sources	127
Annual growth rate (%)	56.51
Citation Information	
Average document age (year)	1.89
Average citations/documents	12.04
Authors Information	
No of authors	688
Authors of single-authored documents	10
Number of co-authors/document	3.42
International co-authorship (%)	0.995

to both databases, 351 unique articles were obtained. The abstracts of these 351 articles were read and the relevance of the articles were judged by the authors individually. Finally, 201 relevant articles were selected based on consensus. The list of these 201 articles is provided in Table A1 in the Annexure. R-software and VOSviewer were employed to obtain the bibliometric and network analysis, respectively.

3 Bibliometric and Network Analyses

The Figure 2 shows the year-wise publication trends of articles. An increasing trend has been observed over the years implying the rapid growth of DL applications in textile operations and quality. Some other important bibliometric parameters are listed in Table 1. The average article age is less than 2 years and the annual growth rate of around 57% which indicates that this research area is at its nascent stage and growing at a very fast pace. Citation details are presented in the form of “average citations per document” (12.04) which shows that articles published in this domain receive quick attention in the research fraternity. Besides, the average number of authors per article is 3.42 with only 10 single-authored articles.

The Figure 3 shows the keyword co-occurrence. Only those indexed keywords that have five or more occurrences were chosen for analysis. Thus 85 keywords were obtained and a minimum cluster size of 8 was chosen. Figure 3 depicts the four keyword clusters created by VOSviewer. Cluster 1 (red) named “DL algorithms” has 24 keywords. “Deep learning” is the central keyword in this cluster with 84 links and 912 link strength. Cluster 2 (green) has 23 keywords, and it is named “Textile applications”. “Textiles” is the most connected keyword in this cluster with 70 links and 231 link strength. Cluster 3 shown in blue has 20 keywords and it is named “Fabric defect identification”. With 64 links and 391 link strengths, “Defects” is the most connected keyword in this cluster. Cluster 4 (yellow) has 8 keywords, and it is named “Neural networks”. “Convolutional neural networks” is the most connected keyword in this cluster with 75 links and 323 link strength. This keyword co-occurrence helps to understand the thematic patterns of the reviewed articles.

4 Content Analysis

4.1 Overview of Deep Learning Techniques

DL is an approach that uses machines to imitate humans to gather particular types of information in order to develop contextual knowledge [61, 130]. To put it briefly, DL is a subset of ML and the latter is a subset of AI [100]. Using the available data, DL trains itself to perform incremental and consistent actions [93]. As a result, it requires less human intervention to create customised, task-oriented software. DL algorithms use multi-layered neural networks functioning on pattern recognition and are further classified into unsupervised, semi-supervised, supervised, and reinforced learning. Unsupervised learning does not require labeled training data, but supervised learning uses the labeled training data to derive a function between input and output. In situations where the relationship between input and output parameters is unknown, unsupervised learning is preferred. Moreover, in cases where data labeling is costly, only a small percentage of the data is labeled and the remainder remains unlabelled. This kind of learning is referred to as

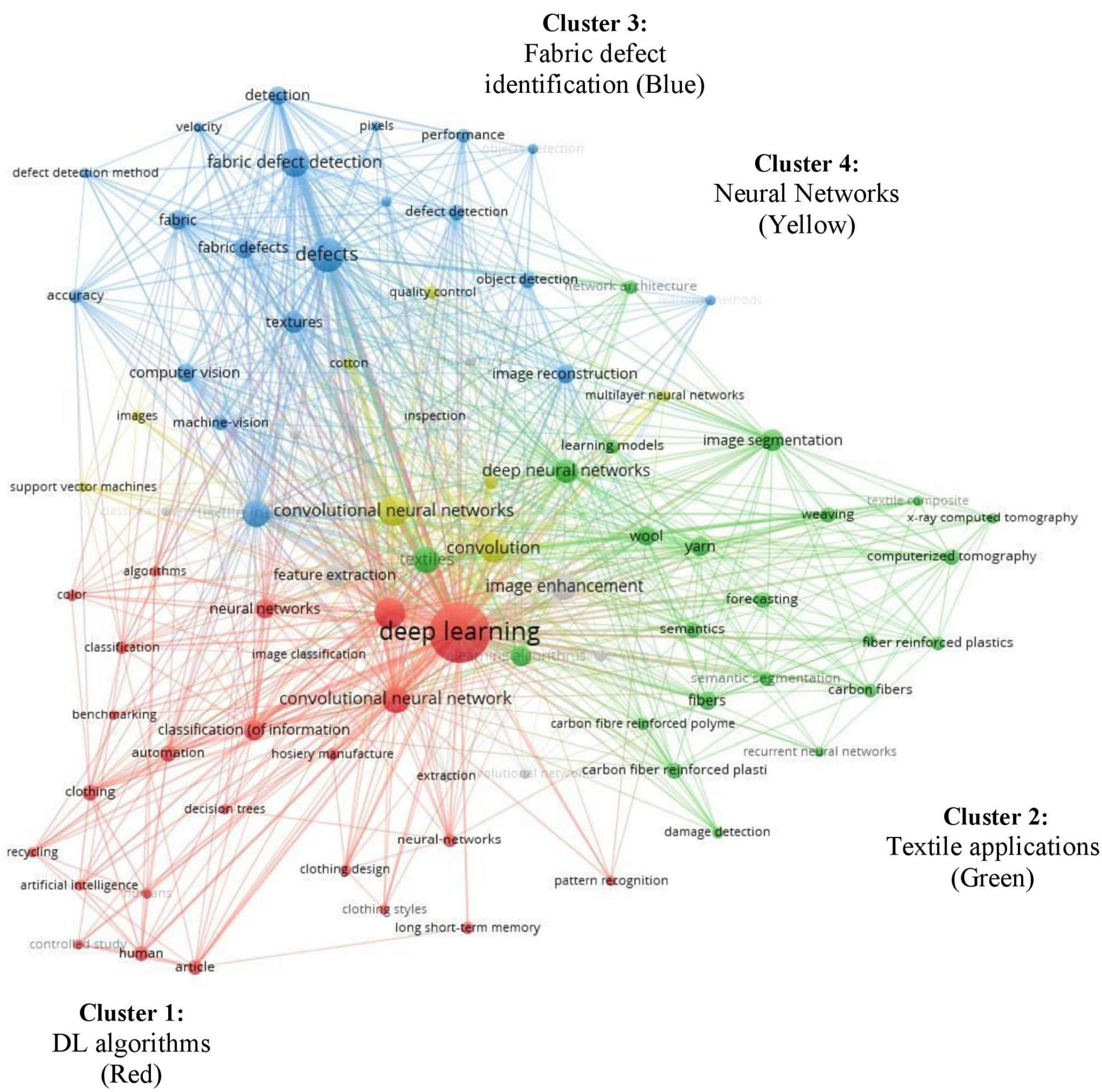

Fig. 3 Keyword co-occurrence network

semi-supervised learning. Reinforced learning involves performing actions based on observations of the environment, which can have positive or negative outcomes. The model is then updated to reflect the decision made about the reward for the chosen action.

In the literature, several DL architectures are discussed and some of the significant ones are shown in Fig. 4. The feed-forward neural network (FNN) is the most basic and earliest form of DL neural network. It consists of several elementary neurons connected in layers [89] . Here the information flow happens in one direction. The recurrent neural network (RNN) utilises its internal state (memory) to remember the inputs given to a node in the previous cycle. Thus, RNN can perceive the sequential characteristics of inflow information and make use of patterns to estimate the next probable event. Popular applications of RNN include natural language processing (NLP) and speech recognition. Long short-term memory (LSTM) works on the same principle as RNN but has a feedback mechanism and can process not only images but also complex audio and video files. Gated recurrent unit (GRU) is an updated RNN version whose capability is almost at par with LSTM. LSTM

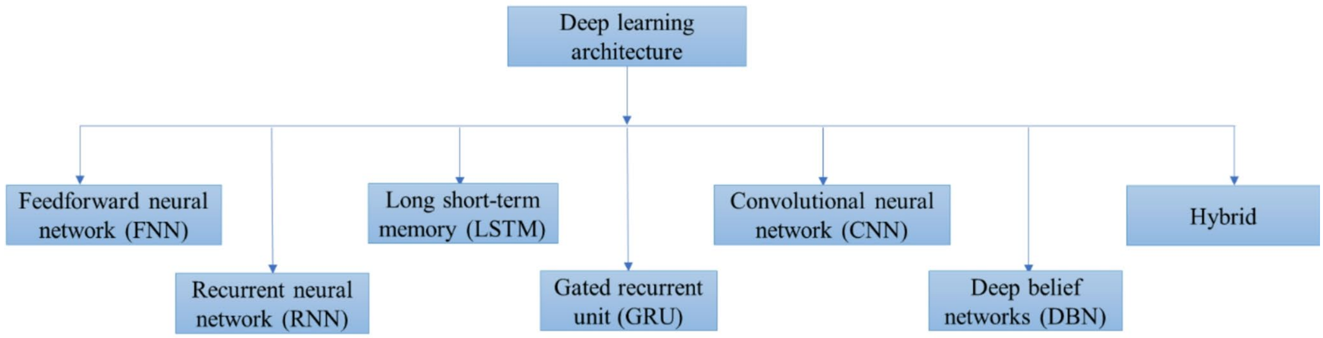

Fig. 4 Common DL architectures

and GRU both improve the capability of RNN to handle long-term dependencies. A convolutional Neural Network (CNN) is a DL architecture that considers an input image and assigns weightage to it in different aspects distinguishing one from another. In CNN, the pre-processing requirement is lesser than other classification algorithms. CNN can learn those characteristics in primitive method filters which are substantially trained hand-engineered. Deep belief networks (DBNs) employ a multilayer network consisting of restricted Boltzmann machines (RBMs) which are nothing but stochastic ANN. Hybrid DL algorithms make use of two or more DL algorithms.

4.2 DL Techniques Applied To Operations and Quality Management in Textile Industry

This section attempts to answer *RQ* 1. Figure 5 depicts the usage of various DL techniques textile operations and quality management. By far the most common DL technique used (36.8%) in the textile industry is the CNN. It was useful in many contexts, from fibre to fabric to logistics and supply chains. When considering hybrid DL systems, which have been used in around 29% of the research articles, CNN is also the most frequently used individual technique. In addition to CNN and hybrid DL systems, “deep neural network (DNN)” and “You Only Look Once (YOLO)” have been used in 18.7% and 8.3% of the articles, respectively, with very little use of the other DL techniques.

Other DL techniques that have found limited application include “partition embedding network” [22], “generative adversarial restoration neural network (GARNN)” [94], “macro-micro adversarial network (MMAN)”, "neural graph filtering collocation framework" [62], “Multilevel and multi-attentional convolutional feature framework (MLMA-Net)” [115], “nested autoencoder” [80], “rule directed pixel comparison (RDPC)” [113], “variable-scale defocusing” [122] “self-feature comparison (SFC) architecture” [81], “skinned multi-person linear (SMPL) model” [29], RNN [64], and LSTM [58, 105].

Since most of the applications are either CNN or its hybrids, in the following sub-section we have discussed CNN and its extensions in detail. The summary of DL techniques at various stages of textile operations and quality management is shown in Table 2. Most of the applications revolve around fabric manufacturing and defect analysis (85), apparel production and classification (53), and fibre identification (51) while yarn manufacturing and quality analysis (11) and supply chain (1) have received scant attention.

4.2.1 Convolutional Neural Network (CNN)

There are a variety of DL architectures that do not require human feature extraction, such as CNN. A CNN is a type of ANN used primarily for image recognition and processing due to the former’s ability to recognise patterns in images. A CNN is composed of an input layer, an output layer, and many hidden layers in between. Three of the most common layers are convolution, activation or rectified linear unit (ReLU), and pooling as shown in Fig. 6. Convolution puts the input images through a set of convolutional filters, each of which activates certain features from the images. ReLU allows for faster and more effective training by mapping negative values to zero and maintaining positive values. This is sometimes referred to as activation because only the activated features are carried forward into the next layer. Pooling simplifies the output by performing nonlinear down sampling, reducing the number of parameters that the network needs to learn. Thus, CNNs can improve the design of traditional ANNs. Optimal parameters for meaningful output and model complexity are taken into consideration in each CNN layer. It is also possible to cope with the problem of over-fitting in traditional networks by using a ‘dropout’. For this reason, CNNs are employed across various ranges of image processing tasks, such as object detection, picture analysis, image segmentation, and many more.

There are enormous applications of CNN in several stages of textiles, i.e., in the fibre stage by [48, 67, 83, 92,

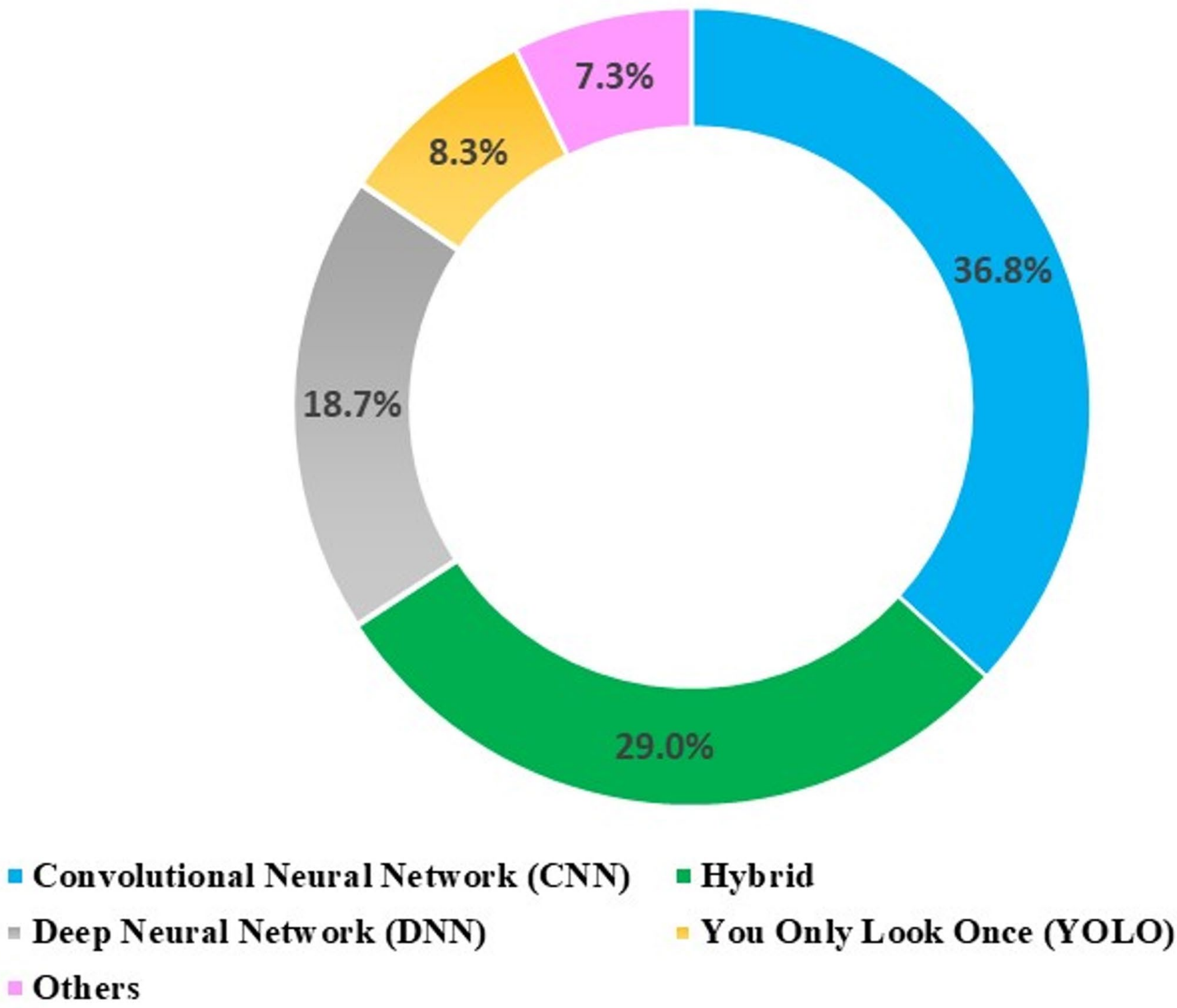

Fig. 5 Usage of different DL techniques in the textile industry

Table 2 Applications of DL techniques in operations and quality management in the textile industry

DL techniques	Process stage in textile operations and quality management					
	Fibre	Yarn	Fabric	Apparel	Supply chain	Total
CNN	11	1	28	17		57
Hybrid	6	1	7	4		18
DNN	16	3	7	14		40
YOLO	2	1	10	3		16
Others (EIT, GARNN, MMAN, MLMA-net, Nested Autoencoder, Partition Embedding Network, RNN, SFC architecture, SMPL model, Variable-scale defocusing, etc.)	16	5	31	14	1	67
Review article			2	1		3
Total	**51**	**11**	**85**	**53**	**1**	**201**

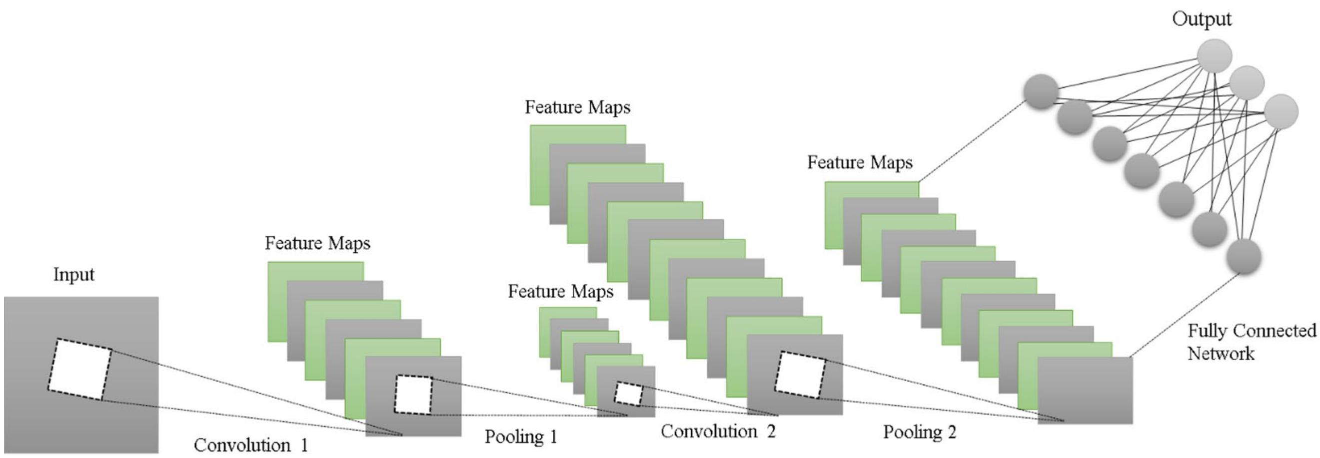

Fig. 6 CNN architecture

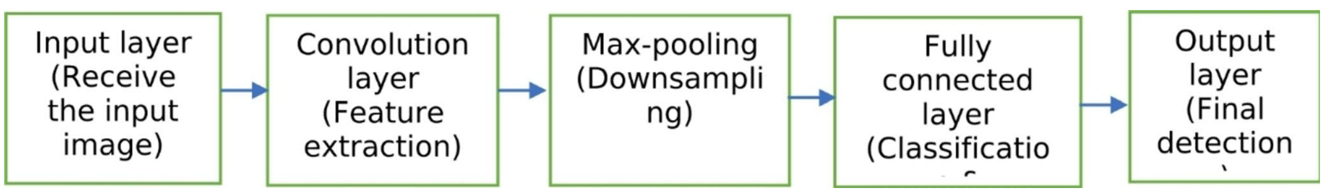

Fig. 7 A block diagram of YOLO architecture

94]; in the yarn stage by Mendoza et al. [73]; in the fabric stage by [34, 37, 38, 42, 79, 95]; in the apparel stage by [9, 15, 18, 128]. Textile supply chain traceability has also witnessed the application of CNN model by using pattern recognition and robust DL [109].

A CNN-based system was developed by Ouyang et al. [79] for on-loom fabric defect identification. In this research, pairwise-potential activation layers were added to a CNN to improve defect detection for complex fabrics with imbalanced data sets. Another algorithm to detect the type of garment and two gripping points that allow the robot to place the garment in a known position was developed by Corona et al. [15]. This system used three CNN-based algorithms to get hold of the garment from the desired gripping points in one drive. CNN has also been shown to be useful in clothing search for video advertisements, as reported in a study on linking sitcom stars to internet retailers via clothing search using deep CNN models called DeepLink [128]. Furthermore, for fabric defect segmentation, Mobile-Unet was utilised by Jing et al. [42]. Compared to U-net and SegN, their approach required five times fewer parameters while achieving higher segmentation accuracy. This model can be trained using an improved loss function even in cases of sample imbalance (i.e., fewer faulty samples).

A semi-supervised deep CNN architecture integrated with a multitasking mean teacher for defect detection was used by Shao et al. [95]. This approach saved a significant amount of cost and labour for labeling the data. The proposed approach outperformed the mainstream methods in terms of evolution metrics while testing on three publicly available defect datasets. Bedeli et al. [9] demonstrated the use of deep CNN for forensic purposes to classify apparel and attributes. Their proposed approach achieved an accuracy of over 70% for low-resolution images. Similarly, Donati et al. [18] developed a deep CNN and image processing method for automatically classifying and recognizing colours, strips, logos, and other clothing features from the rendering images. Their model was tested on datasets of Adidas AG™.

4.2.2 You only Look once

You Only Look Once (YOLO) is a state-of-the-art, real-time object detection algorithm introduced by Redmon et al. [86]. Here object detection problem is framed as a regression problem instead of a classification task by spatially separating bounding boxes and associating probabilities to each of the detected images using a single CNN. YOLO is a faster single-stage detector that processes the entire image at once, while CNNs are two-stage detectors with separate region proposal and object detection stages. Optimisations can be made at every step of the detection pipeline, as the network is a single unit [86]. A simple block diagram of the YOLO algorithm is shown in Fig. 7. YOLO algorithm has found its applications in fabric classification [46], fabric defect detection [41, 60], and fabric wrapping [113]. Some of the improved versions of the YOLO series are: You Only Look Once, Version 3 (YOLOv3), YOLOv5, YOLOv7, YOLOv8, etc. YOLOv3 is a real-time object detection algorithm that recognizes specific objects in videos, live feeds, or images. However, YOLOv3 has some drawbacks, including insufficient small target detection, slower detection speed, and high computational load in fabric defect detection, which calls for further development and optimization [65]. Similarly, YOLOv5 an improved version of YOLOv3, provides higher detection accuracy, faster detection speed, and wider application range. Hu et al. [32] employed the YOLOv5-recursive CNN model for evaluating the detection performance of the generated fabric defect and non-defect images. The network structure and parameter design of YOLOv7 are more optimised, which further improves the efficiency and speed of fabric fault identification. YOLOv8, the computer vision (CV) model architecture is the most recent and advanced algorithm in the YOLO series, resulting in improved feature extraction and object detection performance. This algorithm includes four network architectures involving nano, small, medium, large, and extra-large represented by YOLOv8n, YOLOv8m, YOLOv8l, and YOLOv8x, respectively for several types of applications.

4.2.3 Hybrid DL Models

Hybrid deep learning (DL) models combine many neural network architectures or models into one cohesive system that aims at improving the accuracy of results in models. To handle difficult tasks, hybrid models are developed by combining different neural network architectures, such as CNNs, RNNs, or others, within a single system. There are many hybrid DL models based on the integration of various fundamental generative or discriminative models [91]. CNN has been used extensively with other DL techniques to form a hybrid system. Pre-trained bvlc_reference_caffenet neural network was used for recognising knitted fabric structures (Z. Xiao, Liu, Wu, Geng, Sun, Zhang, & Tong, 2018). The results demonstrated that the suggested technique can overcome the influence of image capture fabric rotation,

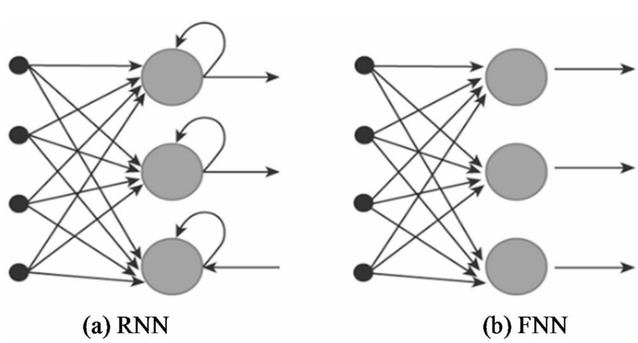

Fig. 8 Recurrent Neural Network (RNN) versus Feedforward Neural Network (FNN)

fabric thickness, fabric hairiness, diameter variation, and light variation and has a high identification rate at a specific time. Additionally, CNN, Field Programmable Gate Array (FPGA), and Digital Signal Processor (DSP) based embedded system was used by Wei et al. [117] for foreign fibre identification and removal from cotton fibre bulk. CNN coupled with edge computing and image processing was carried out by Zhu et al. [134] for fabric defect detection and their method achieved average improvements in the area under the curve (AUC) metric for 11 defects of 18%. Additionally, the method helped to reduce data transmission by 50%, and latency by 32%. Furthermore, numerous hybrid models such as CNN and Random Forest [133], and DL back-propagation neural network model with fuzzy linguistic method [87], etc. have been used for fibre classification. Similarly, the hybrid structure of Faster R-CNN and Feature Pyramid Network (FPN) was used for foreign metal object detection [23]. Deep CNN and watershed segmentation technique was utilised for yarn structure estimation by Ali et al. [4]. CNN, including autoencoder, variational automatic encoder, and generative adversarial networks (GANs) was also found to be efficacious for real-time fabric defect detection [116].

Regular neural networks and recurrent neural networks (RNNs) function somewhat differently. Information in a neural network moves from input to output in a single direction. On the other hand, after every step in an RNN, data is supplied back into the system. By feeding the output from one phase into the next, RNNs enable the network to recollect previous information. This aids the network in comprehending the background of past events and improving its forecasts accordingly. RNNs differ from feed forward neural networks (FNNs) in using loops to allow data from earlier stages to be fed back into the network while the latter do not have memory and thus face trouble in processing sequential input. This difference in their functioning is represented pictorially in Fig. 8. Additionally, deep belief network (DBN) and long short-term memory (LSTM) are shown in Fig. 9 while gated recurrent unit (GRU) is shown in Fig. 10.

4.2.4 Other DL Techniques

Electrical Impedance Tomography (EIT) an imaging technique that reconstructs images of a specific area in the human body was used in the study by Duan et al. [20]. Here, lightweight and low-cost materials having stretch-dependent conductivity were used as artificial skin which was studied for dynamic touch sensing. Apart from the extensive healthcare applications of EIT, researchers explore the prospect of printing a wearable bioimpedance sensor on textile substrates for EIT imaging. Jose et al. [43] developed a flexible textile sensor in the form of a bracelet to obtain

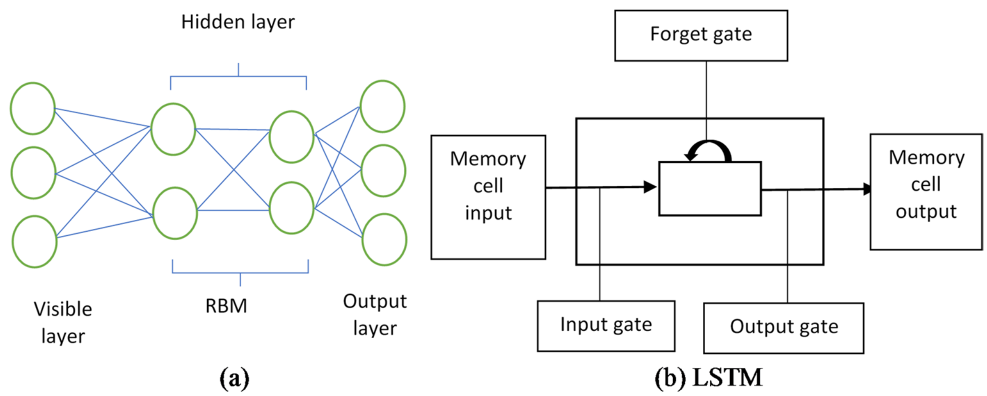

Fig. 9 A simple architecture of DBN and block diagram of LSTM

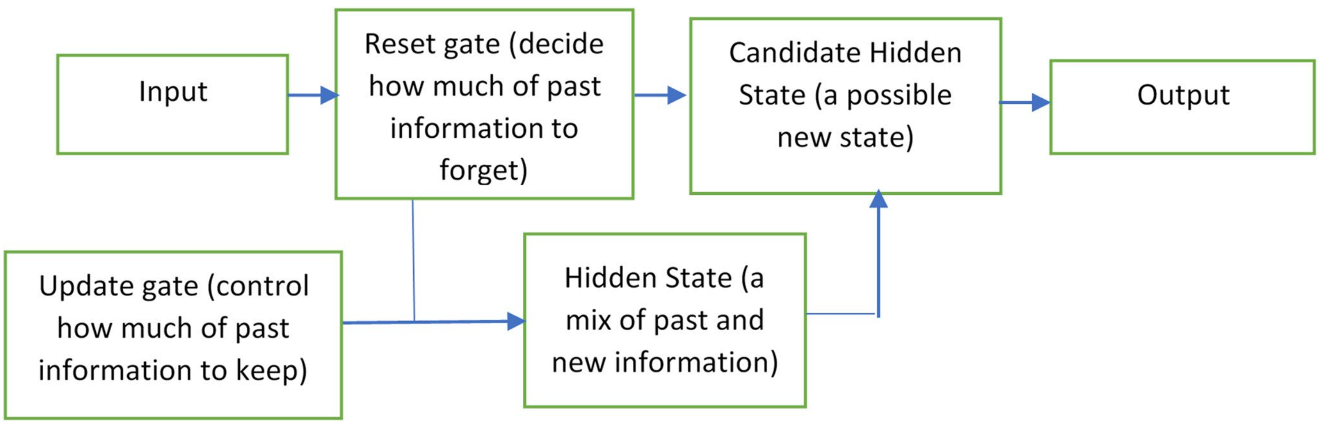

Fig. 10 A block diagram of GRU

cross-sectional images of the forearm that unravel bone features like shape, size, and position. Previous research has indicated that graph attentive recurrent neural networks (GARNNs), a combination of graph attention networks (GATs) with RNNs, can be used to eliminate honeycomb patterns and enhance the spatial resolution of a dual-sensor image to get aligned ground truth data for fibre bundle pictures [94]. Multiple multi-head attentions network (MMAN) was found useful for body shape reconstruction. The original image was first analysed using the MMAN based on DL, and then the results were pre-processed. Sophisticated and handy fashion items like, footwear, bags, etc. were designed using a graph neural network (GNN) [62]. Alaziz et al. [2] developed a 'fashion recommendation system' by applying a Vision Transformer (ViT) having two layers of a multi-head self-attention layer and a multilayer perceptron (MLP) layer to enhance fashion products' performance.

Another interesting framework includes a multi-level, multi-attentional deep learning network (MLMA-Net), which is used in the textile sector to achieve realistic multilevel object recognition of defect images. Features in the foreground and background regions were enhanced and diminished by the multi-attentional module, which is developed to extract high-resolution feature maps from several levels extracted using the multilevel structure [115]. Oz et al. [80] employed nested autoencoders to detect fabric faults since they operate quickly and take up little space. This allowed them to identify flaws online. Furthermore, a study on apparel using a partition neural network was carried out by Feng et al. [22] for garment classification. The two main components of the network were an attribute partition module (which employs numerous labels to guarantee that different regions of the overall embedding correspond to distinct characteristics) and a partition adversarial module (which employs adversarial operations to obtain the independence of distinct parts). Results showed that the embedding has unmistakable portions that represent various traits, and outfits advised by this model are more desirable than those recommended by the existing techniques [22]. Lu et al. [64] used RNN for lace defect detection. To perform defect inspection utilising reconstruction error and a pre-specified threshold (discovered by model training using defect-free samples), a three-step procedure (pre-processing of video, reconstruction of pixel, and classification of pixel) was used. Literature also exists to study the SFC architecture-based fabric defect detection system using fabric images for detection [81]. In this, self-feature reconstruction and self-feature distillation, modules are present, which help in finding defect locations and carrying out segmentation. Skinned multi-person linear (SMPL) model to generate a 3D body shape of the human body utilizing two-depth photos from a commodity depth camera was also attempted [29]. This method was even able to compute the body shape under clothing including pose variations between the shots.

4.3 DL Applications in Operations and Quality Management at Various Stages of the Textile Industry

This section attempts to answer *RQ 2* by presenting the applications of DL techniques at various stages of textile operations and quality management. Most of the applications are focussed on fabric (85), apparel (53), fibre identification (51), while very few applications are seen in yarn (11) and supply chain (1) as shown in Table 2. The summary of stage-wise applications is depicted in Fig. 11. A major share of DL techniques applications amounting to almost 70% have been observed for the textile fabric quality and defect evaluation [34, 42, 56, 79, 85, 134] and apparel [9, 15, 18, 109, 128]. DL application in textile yarn stages has received very little attention. The research works of Ali et al. [4], Mendoza et al. [73], Xu et al. [122], Xu et al. [123] on the yarn manufacturing stage are some of the few pieces of extant literature. Additionally, operations [98], supply chain logistics [109], and forecasting [58] research have received sporadic attention, therefore, there is room for DL to expand its reach in these areas.

4.3.1 DL Applications in Fibre Manufacturing Operations and Quality Management

Understanding the structure and properties of different types of textile fibres makes it easier for producers and clothing designers to determine which type of fibre to manufacture and use. Therefore, precise fibre identification and its classification have become imperative these days and cannot be merely handled by the existing manual methods due to constraints like time, cost, and accuracy. So, to maintain the standards and quality of the merchandise, textile manufacturers can classify fibres and identify their faults through digital image processing and AI applications. In fibre stage, CNN has been primarily used along with other hybrid DL techniques for most of the applications. CNN and GARNN have found applications in fibre bundle imaging [94]. Additionally, CNN has been employed standalone for fibre classification while few hybrid applications for fibre classification are also seen in the extant literature [67, 117, 125, 31]. In recent years, extensive research works on the impact of micro fibres in composites on load-carrying capability, fatigue life, and the effect of defects have been published, aiming to achieve lightweight structures of aircraft, ships, and automobiles. To this end research studies of [17, 25, 33, 107] are noteworthy.

DL algorithm was used to evaluate the quantity of foreign fibre in raw cotton [118]. They used CNN architecture

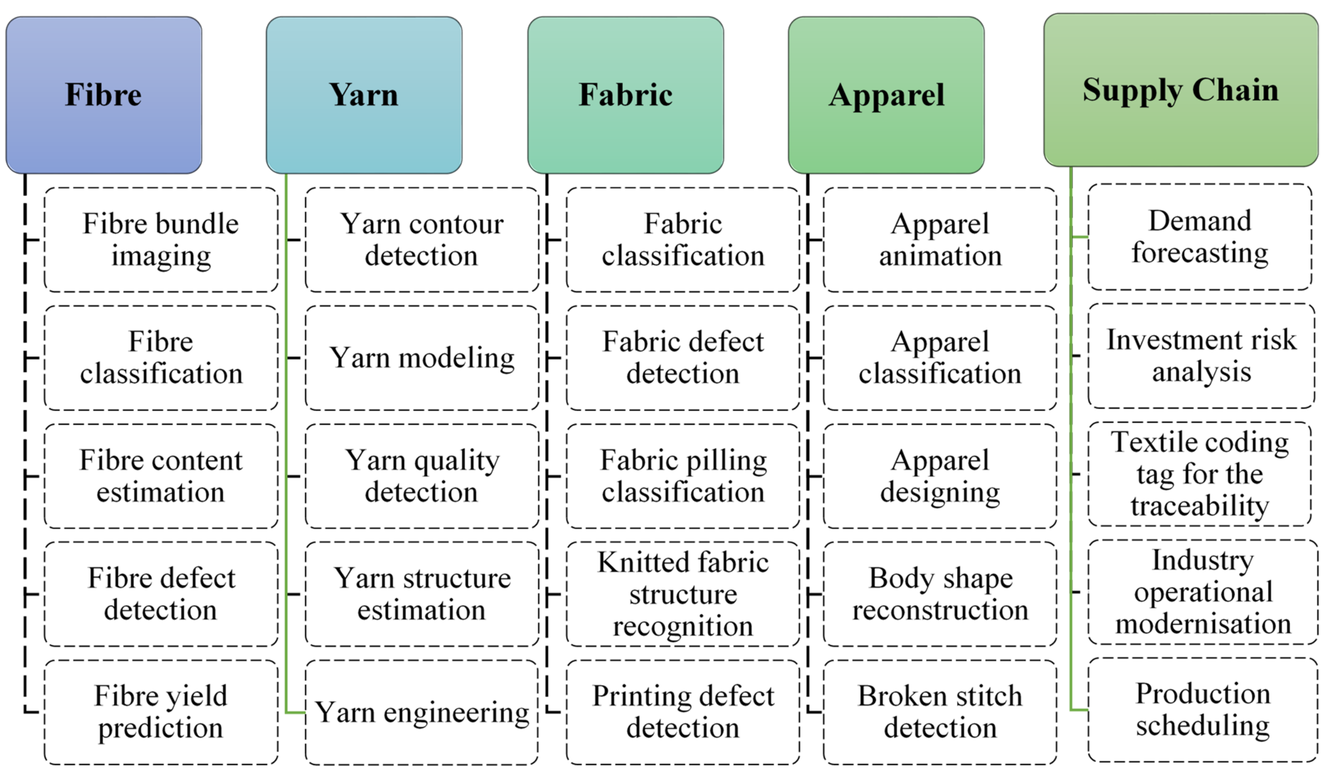

Fig. 11 Major applications of DL in operations and quality management in the textile industry

based on U-Net in their study. The results demonstrated that there was less than a 4% relative error between the estimated and real sizes of the foreign fibres and the overall error was found to be less than 2% by comparing the algorithm data collected at random with the statistical data from the real foreign fibre. In earlier work, an embedded system based on a field programmable gate array (FPGA) combined with a digital signal processor (DSP) to recognise and remove foreign fibres mixed in cotton was proposed [117]. The developed model achieved a success rate of 96% corroborating its effectiveness. An object detection and classification algorithm using an improved YOLOv5 to detect and classify small cotton foreign fibres was developed by Wang et al. [110]. The method improved the detection speed and model volume, enabling application in production line image processing systems. With a huge dataset of 13,600 leaf photos from 27 different cotton genotypes a DNN architecture was able to classify the leaf images [88]. For both the image and the leaf, the model showed an accuracy of 89% and 95%, respectively.

Another interesting problem where DL has found interesting applications is the identification of wool, mohair, and cashmere fibres. Though all these fibres are broadly classified as wool, they originate from sheep, Angora goats, and Kashmir goats respectively creating differences not only in their scale patterns but also in thermal properties. A techno-economically viable method to separate wool fibre from mohair by using a DL-based texture analysis identification model was developed [125]. Pre-processed micrographs of mohair and wool fibres served as the texture images. To extract conclusive information from the fibres, two different methods were utilised: DL and a local binary pattern-based feature extraction procedure. It was possible to detect wool and mohair fibres individually with an accuracy rate of 99.8% and 90.25%, respectively. On the other hand, a hybrid model based on CNN and Random Forest for automatic feature extraction and classification of wool and cashmere fibre micrographs was also explored [31]. The results of the investigation showed that compared to other classification techniques, the suggested method performed better and had a greater classification accuracy (Fig. 12). A rapid qualitative wool content assessment method using near-infrared spectroscopy (NIR) and U-Net++, achieving 93–94% accuracy, recall, and precision was also developed recently [55].

UNet3D-based CNN architecture to compare existing segmentation algorithms with the performance of CNNs was developed by Kibleur et al. [48]. This model successfully identified the fibres confined in medium-density fibreboard complex networks and accurately estimated the fibre density by morphometrically measuring their bundles. CNN-based Muller matrix M_{44} was used for the classification of electrospun fibres derived from polycaprolactone (PCL) and poly-L-lactic acid (PLLA) polymers for tissue engineering applications [67]. The Muller matrix was calculated using the polarised information of the different fibres that was garnered using a polarised light microscope. This was followed by the application of transfer learning which allowed for the automatic categoris ation of the microporous morphology of electrospun ultrafine fibres leading to an accuracy of 96.15%.

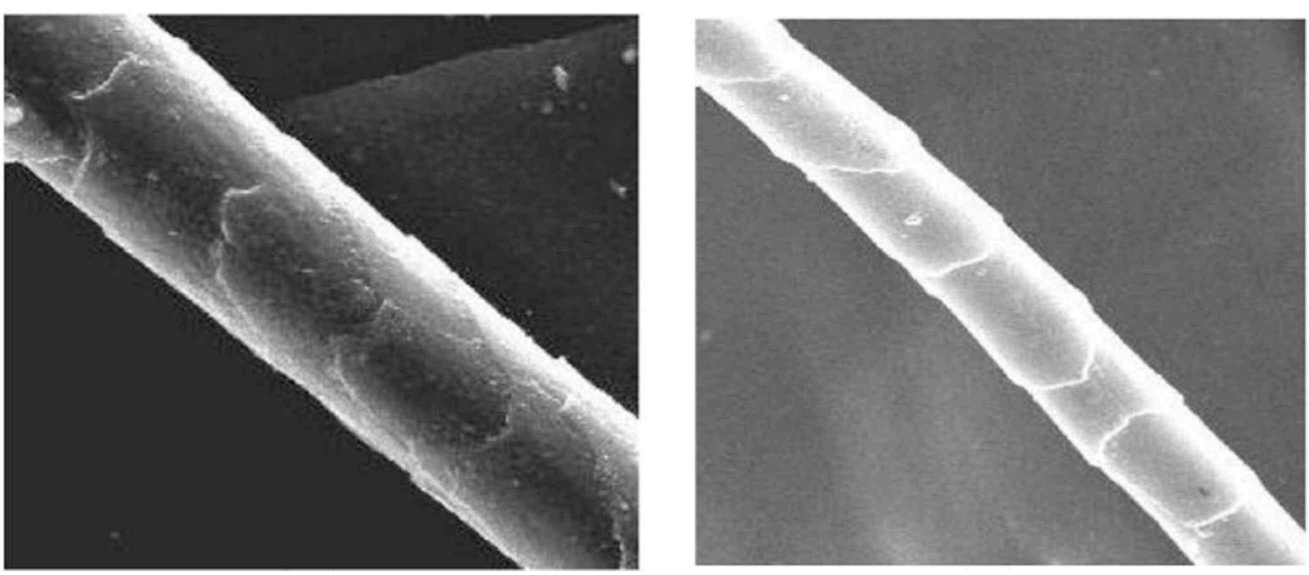

Fig. 12 Images of wool and cashmere fibres (Zhu et al., [133])

4.3.2 DL Applications in Yarn Manufacturing Operations and Quality Management

The applications of DL in yarn manufacturing operations and quality management stage are found to be relatively sparse. A textile yarn's diameter, uniformity and hairiness, twist orientation, number of filaments, thick and thin places, presence of entangled fibres (neps), etc. are some of its characteristics that can affect performance during weaving. Over the years, some researchers have proposed empirical, mathematical, statistical, and practical solutions to address these issues. In recent years, several methods based on ML techniques have also been used to predict yarn quality, geometry, orientation, defects, etc.

A feature subspace model based on multi-correlation parameters for predicting yarn quality variations was proposed by Hu et al. [30]. Primarily they used the Kernel principal component analysis (KPCA) algorithm followed by a deep belief network (DBN) and finally the particle swarm optimisation (PSO) algorithm to optimise the modeling parameters. A completely automated and unique technique for obtaining yarn geometrical features to define parameters that correlate with textile reinforcing in composites was successfully developed [73]. They performed yarn segmentation from computer tomographic (CT) pictures that were precise and effective, as demonstrated by the experimental results and further supported by quantitative and qualitative assessments. Detection of accurate contours of the yarn body from images by employing a remote data possession checking (RDPC) knowledge-based contour detection method with mask layers was attempted by Xu et al. [122] and a series of diameter values to evaluate the yarn evenness was obtained (Fig. 13). Comparative tests conducted during real spinning operations showed that the suggested approach lowered the quality inspection error from around 1.5% to an acceptable 0.53%. In another study, the relationship between the defocus blur degrees and the poses of the yarn body to establish the critical role of image deblurring in vision-based detection systems was investigated [122]. The developed knowledge-augmented CNN for yarn images deblurring with variable-scale defocusing helped to get better input for accurate inspection. Sophisticated features of deep CNN can be used to segment CT scan images of a multi-layered plain-woven fabric from unseen datasets and isolate individual yarns with connected cross-sections during post-processing of segmented volumes [4]. Deep CNN was trained with a manually annotated collection of raw 2D image slices produced from a single-layer fabric's grayscale volume. The developed network managed to generate digital material twins with more than 96% global accuracy. In very recent work, machine learning (ML) to create a semi-supervised anomaly detection technique that uses a convolutional autoencoder (CAE) was explored by Wang et al. [107]. To determine the optimum value for anomaly detection, they employed the reconstruction mode of a CAE and Kernel density estimation (KDE).

4.3.3 DL Applications in Fabric Manufacturing Operations and Quality Management

Among the segments of the textile and clothing value chain, fabric manufacturing operations and quality management have seen the maximum applications of DL. These applications include fabric classification [34, 66, 68], fabric defect detection [37, 38, 42, 90, 126, 56,79, 95, 134], fabric pilling classification [97, 120], fabric surface characterization [14, 31, 39, 51], fabric wrapping [113], knitted fabric structure recognition [121], lace defect detection [57, 64] and printing defect detection [12, 119].

A DL model for the categorisation and recognition of woven fabrics that is based on data augmentation and transfer learning was developed as shown in Fig. 14 [34]. The

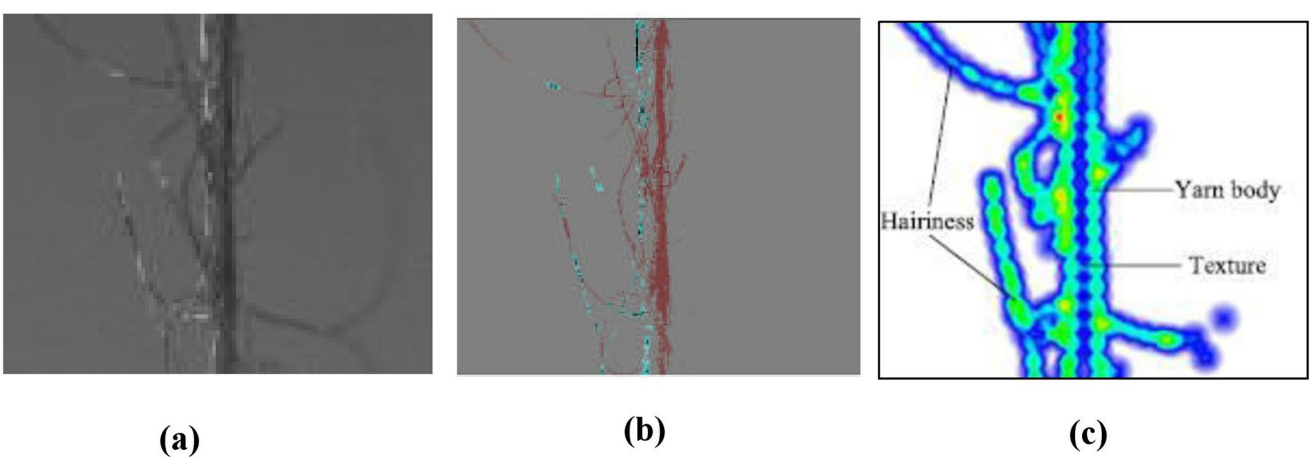

Fig. 13 Vision-based yarn evenness inspection analysis (**a**) Raw yarn structure (**b**) Pixel intensity heatmap of the yarn sample (**c**) Gradient heatmap of the yarn sample (Xu et al., [122])

model took advantage of the residual network (ResNet) architecture, which automatically extracted and classified the fabric texture features end-to-end. The method produced higher accuracy with rotating orientations of fabric. In another research, two CNNs, namely Visual Geometric Group (VGG19) and ResNet50 were used to mechanically recognise and classify ceramic fabrics using petrographic thin section analysis (Lyons et al., [66]). The results showed that classifying ceramic fabrics using CNNs is a very practical and efficient approach.

An image pre-processing technology and DL network for classifying the pilling level of knitted fabrics was also demonstrated [97]. An image optical sensor was used to gather fabric images followed by the use of a DL network to automatically capture and identify the features and classification of fabric pilling. The proposed model was found to be superior to the type-2 fuzzy cerebellar model articulation controller (T2FCMAC) and deep-principal component-analysis-based neural networks (DPCANN), with average accuracy differences of 0.3% and 2.7%, respectively. In another work, an innovative vision for assessing the pilling quality of fabrics was developed [120]. The technique involved converting the images into the frequency domain by a Fourier transform followed by the determination of the optimal degree of subdivision using the multidimensional discrete wavelet transform and an iterative thresholding to obtain a complete and clear picture of the pilling region. Finally, the DL algorithm was used for training and the pilling levels were classified with an accuracy of 94.2%. The fabric wrapping in the field of medical surgery has significant importance [113]. The experiments conducted by the authors demonstrated the efficiency of the developed method to accomplish complex manipulation tasks. The results also showed that the method can be generalized to different colored fabrics and boxes with different sizes, textures, or geometric shapes. A DL-based approach (AlexNet) for knitted structure recognition was introduced by Xiao et al. [121]. The suggested recognition method can obtain a high recognition rate while being fully inert against the effects of uneven light, fabric hairiness, and rotation.

A large volume of research articles focuses on the identification of fabric defects be it woven, printed, or lace fabrics. In very recent research, a deep learning-based fabric inspection system for identifying fabric defects was presented [126]. The method utilised a saliency-based region recognition technique to identify the defective areas in fabric photographs and determine the Region of Interest (RoI). The fabric images were then categorised using a CNN into defective and non-defective images, respectively. The suggested strategy achieved 95.8% average accuracy. Earlier fabric defect detection and classification was also developed by Jeyaraj and Nadar [38] by employing a multi-scaling Deep CNN algorithm. The authors compared its results with the conventional system to validate an accuracy of 96.6% of the developed architecture (Fig. 15). In another work by the same researchers, fabric defects using ResNet512-based CNN were studied resulting in a classification accuracy of 96.5% with 98.5% precision [37]. A novel defect detection model for jacquard-patterned fabrics using CNNs and other DL models with image pre-processing was utilised

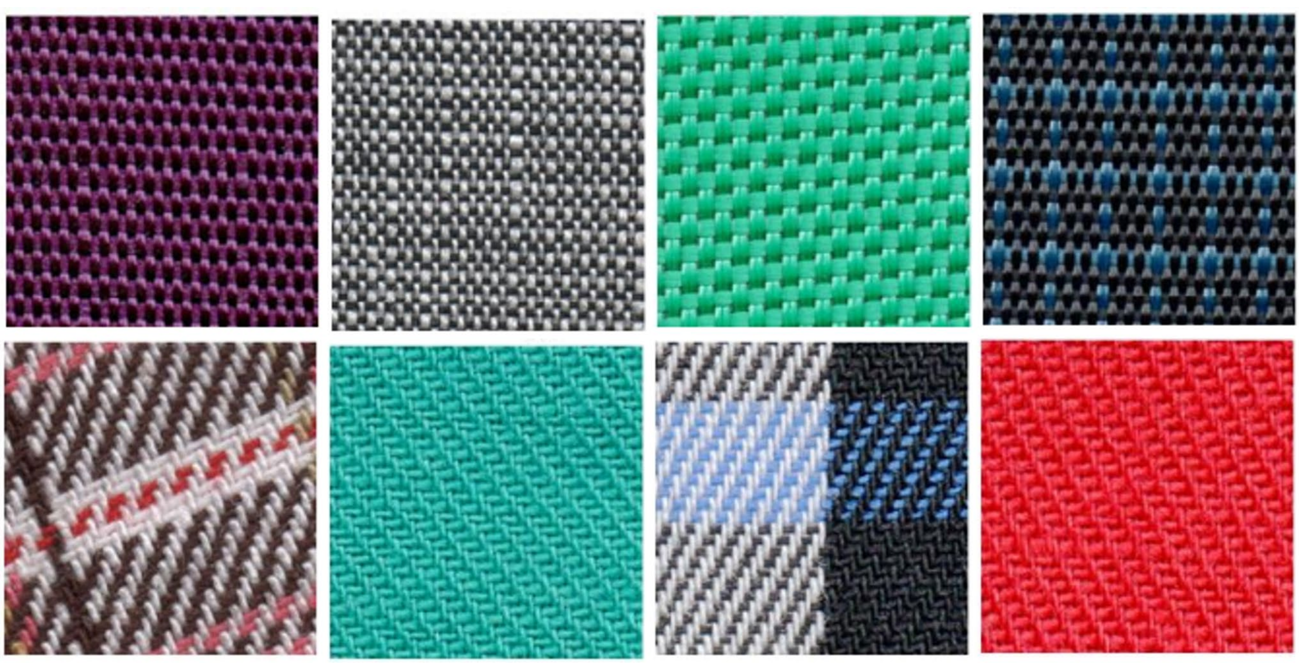

Fig. 14 Identification of plain and twill woven fabrics [34]

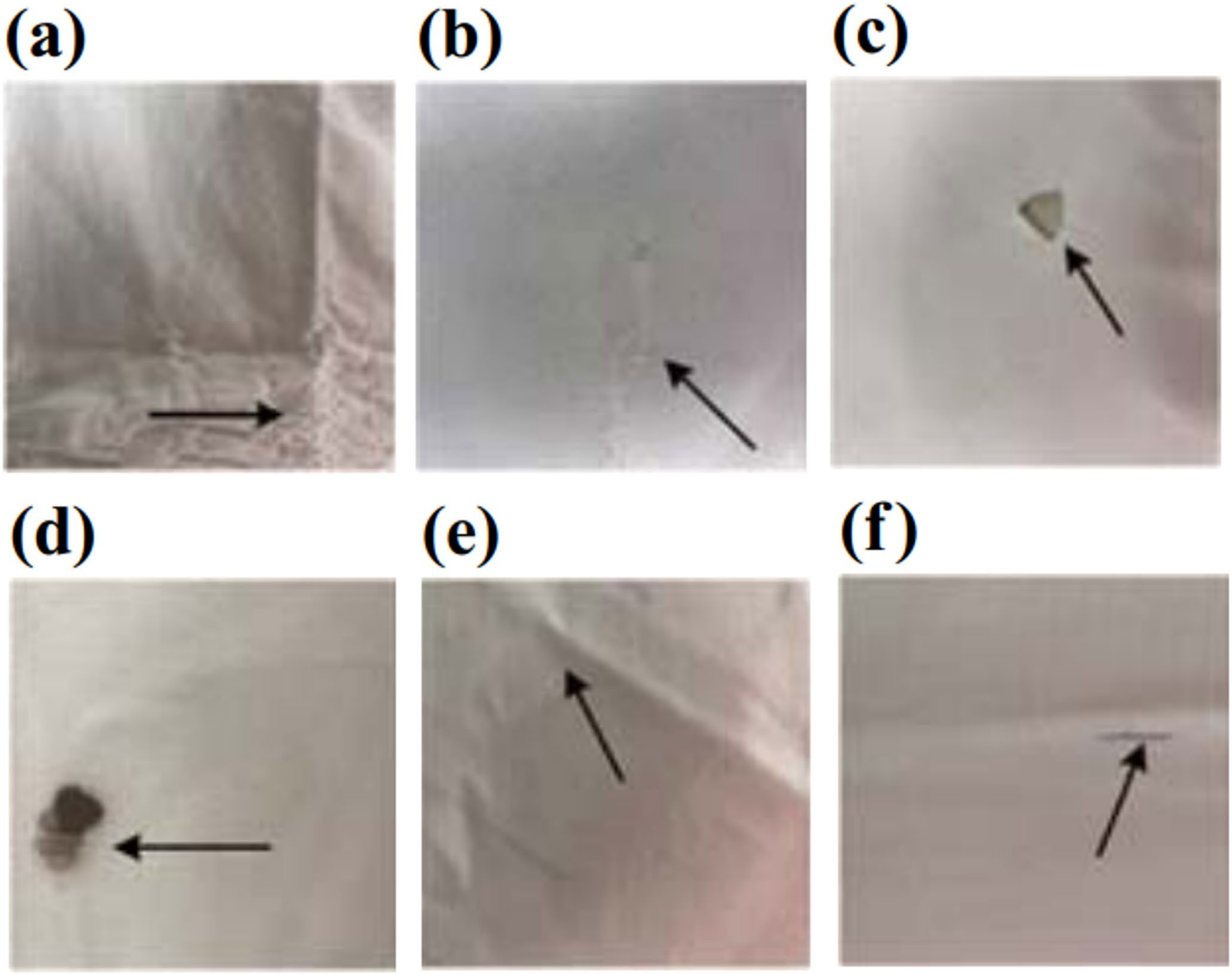

Fig. 15 Different types of defective fabrics **(a)** slub; **(b)** stitching; **(c)** hole; **(d)** rust strain; **(e)** weft curling; **(f)** fibres miss pick [38]

for the unsupervised defect identification [47]. Additionally, near-infrared (NIR) and conventional (RGB) imagery were combined to create multispectral imaging, which helped them enhance and boost the accuracy of their system. Mobile-Unet for endwise fabric defect segmentation was also explored [42]. The performance of the developed model was evaluated by public and self-built fabric datasets and was conferred with better accuracy and detection speed over other existing detection methods. Some researchers advocated Fisher criterion-based stacked denoising autoencoders (FCSDA) framework to detect defects in plain, twill, periodic patterned, and jacquard warp-knitted fabrics [56]. Experimental results established the superior detection accuracy of the developed network. A DL algorithm can also perform fabric defect inspection on a running loom by combining image pre-processing, fabric motif determination, candidate defect map generation, and CNNs [79]. The results showed that the average accuracy of detecting defects was over 90% and 80% at the pixel level, and the precision of counting the number of defects from a publicly available data bank was over 98%. Fabric defect detection can also be addressed by developing a pixel-wise semi-supervised fabric defect detection method in combination with a multitask mean teacher [95]. The authors focused on building a multitask detection model for detecting defect areas, defect contours, and defect distances by utilising their additional information for better defect segmentation. Experiments on three public data pools for defect detection showed that their method outperformed current pixel-wise segmentation methods using MT. In another work in this area, a DL-based fabric defect detection method for edge computing scenarios was suggested [134]. The data set was expanded using six workable expansion schemes based on the features of different types of defects in fabric samples. The optimized model outperformed a CNN by an average of 18% in the area under the curve (AUC) for 11 faults. A novel methodology showcasing the application of CNN to classify printing defects based on fabric images collected in the industry was proposed by Chakraborty et al. [12]. The study additionally integrated DL networks like VGG, DenseNet, Inception, and Xception to compare the performance of the models. VGG-based models outpaced a basic CNN model, suggesting a promising automatic defect detection system for printed fabrics. A system for DL-based irregularity detection to identify fabric lace defects was developed by Lu et al. [64]. The three stages of the framework are the pixel reconstruction stage, the pixel categorisation stage, and the video pre-processing stage. The usefulness of the proposed framework was proved through experimental findings on videos with simulated defects.

4.3.4 DL Applications in Textile Chemical Processing

This subsection discusses the use of in textile dyeing, printing, and finishing, focusing on color prediction, pattern recognition, defect detection, and predicting fabric thermosetting parameters in dyeing, printing, and finishing, etc. This manifold study witnesses ANN, various CNN variants like AlexNet, conventional machine learning techniques like support vector regression, principal component analysis, XGBoost, generative artificial intelligence like generative adversarial networks, and genetic algorithms, etc. Ingle and Jasper [35] have done an in-depth review of several DL applications like back propagation neural network (BPNN), adaptive neuro-fuzzy interference system, ANN, deep Q-network (DQN), forward feed neural network (FFNN), probabilistic neural network (PNN), random vector functional link network (RRVFL), etc. in textile dyeing, bleaching, printing, etc. Solaiman et al. [99] addressed the carcinogenic effects of the industrial dye pollutants present in the textile effluent and tried to mitigate this serious issue by exploring the application of loose nanofiltration (LNF) porous ceramic membranes in treating and recuperating the textile wastewater. They predicted the filtration efficiency of the LNFs by using ANN. Pervez et al. [82] developed a least square support vector regression (LSSVR) model based on Taguchi's statistical orthogonal design (L_{27}) to predict exhaustion percentage (E%), fixation rate (F%), total fixation efficiency (T%), and colour strength (K/S) in the reactive cotton dyeing process. Jiang et al. [40] exploited 'Orange' a free ML neural network building software to predict the pH strip values fostering DL applications in the textile chemistry field.

4.3.5 DL Applications in Apparel Manufacturing Operations and Quality Management

Like the fabric stage, the apparel manufacturing operations and quality management stage of the textiles have also witnessed good penetration of DL applications. In recent years, the incredible increase in online shopping has boosted the applications of image understanding and retrieval, as hundreds of thousands of images of garments represent a voluminous data set that can be used for automatic or semi-automatic colour and texture analyses, similarity retrieval, etc. Some of the notable applications are apparel designing [124, 127, 132], apparel animation [109, 112], and apparel classification [63, 71, 112, 5, 9, 15, 18, 21, 22, 44, 62, 72], apparel detail enhancement [129], apparel colour detection [54], clothing pattern recognition [13, 77, 101, 131], apparel insulation prediction [76], body shape reconstruction [11, 29, 62, 129], broken stitch detection [49, 52, 106], clothing identification from video advertisement [6, 128], foreign metal object detection [23] and even recycled clothing classification [78].

Attention has also been paid to the clothing features to apply DL in clothing design [127]. The images of women's outerwear were selected using a reverse image search. Based on the results of the online survey conducted, it was considered advisable to use specific characteristics of the respective garment as labels and not merely use its name. To this end, a DL-based method for semi-automatic garment animation was also developed [109]. In this method, the user selects keyframes that include the desired garment shape. The methodology was assessed by using typical clothing styles in both qualitative and quantitative ways. The experimental data showed that the method drastically reduced the ratio of required keyframes from 20% to 1–2% when compared to the typical computer-graphics pipeline.

A technique for identifying individuals using the visual signals provided by their clothing was developed by Bedeli et al. [9]. The classification of clothing was based on a dataset that included well-known brand images and logos. The model successfully classified the majority of the test photos with an accuracy greater than 70%. Additionally, an assessment of clothing classification using security camera footage was conducted that gave correct labeling of 70% of the test images. Corona et al. [15] proposed a CNN algorithm that first, identified the type of garment and then performed a search of the two grasping points that allow a robot to bring the garment to a known pose. The results obtained in the three steps (recognition, first grasping point, second grasping point) were found to be quite promising. A partitioned embedding network can learn interpretable embeddings from clothing products and comprehend the significance of many features in an outfit composition for designers,

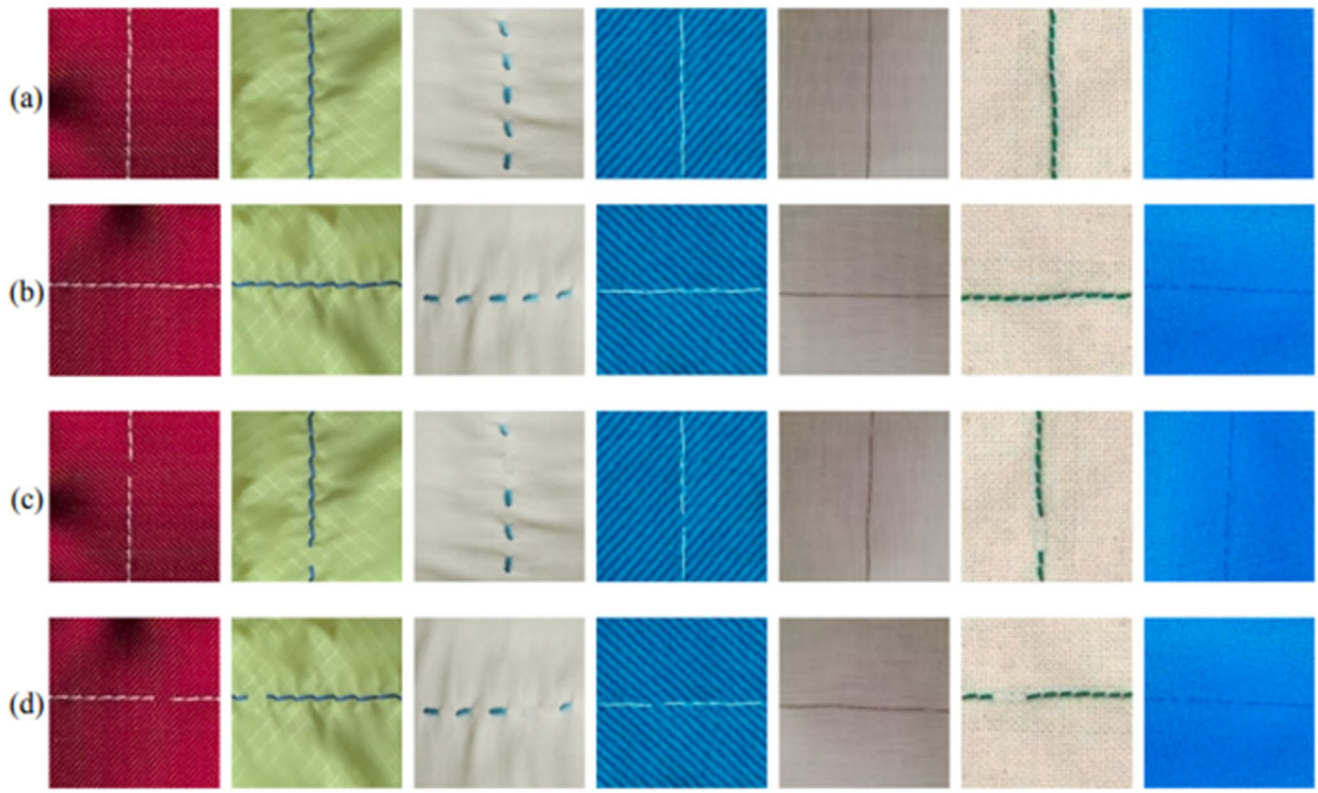

Fig. 16 Broken stitch defect detection in apparel **(a)** original sample; **(b)** rotated 90º clockwise of (**a**); **(c)** synthetic defect of (**a**); **(d)** rotated 90º clockwise of (**c**) [49],

businesses, and customers [22]. In another research, a neural graph filtering framework modeled a flexible set of fashion items via a graph neural network to enable flexible and diverse fashion comparisons [62]. The suggested techniques significantly enhanced the field of fashion studies. In their investigation of real-time clothing garment classifiers, Medina et al. [72] focused on single garment recognition for texture, fabric, shape, or style. They then proposed a CNN model classification for the application of these classifiers on cameras by analysing and contrasting the ResNet18, visual geometry group (VGG16), and Inceptionvv4 classification algorithms with the fashion-modified National Institute of Standards and Technology database.

In recent work, mixed swin transformer U-net (MST-Unet), a semantic segmentation network model, was developed to improve ethnic clothing pattern segmentation [77]. Before this Wang and Sun [112] explored the use of DL in ethnic clothing culture and pattern design by proposing a pattern-generative model using a two-layer LSTM network for decoding. Cloth stiffness values from video scenes for virtual cloth simulation using deep learning models like Transformer can also be predicted [71]. The model used Position-Based Dynamics (PBD) to accurately categorise videos based on softness-to-stiffness labels, enhancing the efficiency of video classification tasks. Kim et al. [49] carried an interesting work using image processing and CNN feature maps to identify sewing defects with 92.3% accuracy (Fig. 16). The method performed well in garment manufacturing factories.

A predictive model for clothing insulation levels of building occupants was developed using DNN [76]. Their work revealed a stronger influence of outdoor environment factors on clothing insulation levels as compared to indoor environment elements. In another research, lean and Six Sigma models in conjunction with the DL technique for boosting the Moroccan textile and clothing companies were attempted [21]. Around 1200 companies were studied and the model was trained to predict the success rate of lean and Six Sigma implementation.

4.3.6 DL Applications in Textile Supply Chain Management

Among all the stages, supply chain management has seen the least penetration of DL. However, DL has significant applications in the textile supply chain leading to its transformation by improving efficiency, optimising processes, and enabling intelligent decision-making. Some of the key applications include quality inspection, counterfeit detection, defect detection, inventory management, demand forecasting, waste reduction, and sustainability compliance, personalised and smart textiles, automated production and robotics, and predictive maintenance. Owing to the availability of huge data along with the demand for accurate forecasting, there lies a tremendous scope for DL penetration in textile supply chain management [109]. dealt with logistics and used faster R-CNN for textile coding tags and coded recognition to have better traceability of the product during operations. Based on the unique optical characteristics of each tag, the tag recognition process traced and classified the individually coded yarns [109]. Another study in this field used a combination of a residual network (ResNet) and LSTM risk prediction models for predicting foreign investment risk in the textile sector [58].

The critical analysis of reviewed articles showed that the application of DL techniques is by and large limited to the fabric and apparel stages. This implies that there is a necessity to take initiatives in a broader spectrum by employing DL techniques from the micro fibre and meso yarn stages to the forecasting and logistics stage in textile industries. Apart from these, there is little to no penetration of DL in textile chemical processing, which is quite a vast area to be explored.

5 Future Research Directions

This section deals with *RQ* 3. There are quite a few segments of the textile value chain where DL penetration is either very low or non-existent. These include, in particular, yarn manufacturing, chemical processing, and supply chain management. Future research should explore DL applications in textile chemical processing as it is a value addition stage and even a small improvement in this can lead to hefty cost savings. Defect detection in fabric has seen vast implementation of DL techniques that range from yarn defects to fabric and printing defects. To achieve even greater efficiency, a unified system for textile inspection will require the addition of an increasing number of defect categories as well as the creation of universal datasets for model training. Thus far, research has demonstrated that greater penetration and acceptability in the textile industry can be achieved with less processing power requirement for DL implementation. In our daily lives, we encounter issues with the actual shade or texture not matching with those on the e-commerce platform. DL can be utilized to match the original product's attributes with the displayed image. Through the use of training-based learning, this can even go further in correcting the displayed picture's shape and colour to match those of the original product. DL has been employed for developing textile coding tags for improved traceability in the supply chain. But presently this is limited to coding of yarns in woven fabrics only and there lies a scope for developing this sort of system for knitted and nonwovens soon, for which the role of embroidery-based coding can be explored.

The textile fibre stage has seen most of the applications of DL in fibre analysis and characterization phase but not in the manufacturing phase. DL holds great promise for enhancing fibre yield, and quality and developing novel manufacturing methods. Improvements in quality are prioritized in the yarn manufacturing stage; however, to achieve the best results in terms of tactile qualities and aesthetic appeal, it is necessary to optimise the parameters of blending or mixing different types of fibre. To this end, DL using raw data can be very beneficial. Even though DL has been used to its fullest extent at the fabric defect identification stage, there is still little to no emphasis on using DL to enhance the textural performance of the fabric. In the textile supply chain, DL's penetration in operations and logistics is minuscule. Therefore, research on predicting future fashion trends as well as data-driven real-time decision-making can be conducted to improve fashion forecasting, ensure sufficient supply to meet generated demand, and minimise loss from inventory reduction.

Some of the textile classification problems are very intricate and yet to be solved conclusively. Therefore, DL should be applied to solve such problems as differentiating original and counterfeit products; or distinguishing fabrics manufactured on handloom and power-loom. These will help protect the brand equity and heritage, respectively. There is also an impending need to find new DL techniques that can be used in places where initial training with large datasets is not possible, and the developed model should be capable of working with limited data. To make all of these changes readily available and then better utilised, transfer learning can be used. Eventually, all of this may result in the textile sector becoming intelligent rather than governed by predetermined rules and perceptions.

6 Conclusions

This work presents a mixed review approach comprising bibliometric, network, and content analyses to understand the current status and future research directions of DL applications in operations and quality management in the textile industry. Among the textile manufacturing stages, most of the applications of DL revolve around fabric and apparel stages (fabric and apparel classification, structure determination, fabric defect detection, body shape reconstruction, etc.). CNN and hybrid DL systems are leading the way in the broader use of DL techniques. However, DL has very slightly permeated at the upstream side of the textile value chain, such as fibre and yarn manufacturing. Furthermore, there are very limited applications for DL in forecasting, supply chain, and logistics of the textile industry. Therefore, more collaborative and interdisciplinary research is needed to create an intelligent, responsive, and efficient textile operations and supply chain. By classifying DL applications in the area of textile and clothing and outlining its future research directions, this study contributes to the current body of literature by creating a knowledge map. Besides, by analysing the entire textile value chain, from fibre production to supply chain, this study offers a bird's eye view that will help the researchers and practitioners working at the interface of DL and the textile industry.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose. The authors have no competing interests to declare that are relevant to the content of this article. All authors certify that they have no affiliations with or involvement in any organisation or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no financial or proprietary.

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